

The Information Isomorphism Law: A Meta-Law Spanning Ontology and Epistemology

——Tracing the Necessary Premise from the Cross-Disciplinary Success of the Scientific Discovery Operating System

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Abstract

The Scientific Discovery Operating System (Tang, 2026g, 2026r) has been successfully deployed across six completely independent disciplines, yet a fundamental question has never been systematically addressed: Why is this operating system effective? This paper's answer is: because in complex systems grounded in objective data and exhibiting regime switching, statistical structures carry the information of regime structures. This ontological fact is formally named herein as the Information Isomorphism Law. The argumentative path of this paper is not induction from specific discoveries to a universal principle, but a retracing from the operating system's success to its necessary premise: the six functional dimensions of the Testability Norms—Search Completeness, Modulation Relationship, Structural Breakpoints, Causal Direction, Problem Completeness, and Process Closure—together constitute a multi-dimensional probe of statistical structures. If statistical structures contained no information traces of regime switching, this methodology could not have succeeded repeatedly across six disciplines. Therefore, the success of the operating system itself proves the establishment of the Information Isomorphism Law. The Chow F-order parameter duality (Tang, 2026k) and the intercept/slope ratio-critical distance duality (Tang, 2026x, Theorem A) provide precision verification for this law on a truth-known system, rather than serving as its inductive basis. The Information Isomorphism Law is an ontological law—it asserts a fundamental fact about how the world is organized: "Statistical structures carry the information of regime structures." Precisely because it describes the special fact of "information carrying," it is naturally and simultaneously an epistemological meta-law—it answers the fundamental question of "how we can know the world." The Factor Hierarchy Law (likewise a meta-law) simultaneously revealed, at the moment of its discovery, both an organizational principle of the world and the statistical visibility of that principle—it inherently carries an epistemological dimension. The Information Isomorphism Law is the universalized extraction of this original unity—it separates "visibility" from the specific content of the Factor Hierarchy Law and establishes it as a universal law applicable to all regime switching across all statistical dimensions. The two laws are complementary, jointly constituting the complete philosophical foundation of the Scientific Discovery Operating System. Within the operating system, the Information Isomorphism Law occupies the epistemological foundation layer, completing a comprehensive four-layer structure. It must be candidly acknowledged that the Information Isomorphism Law was explicitly proposed only after the completion of the preceding 28 papers—historically, the operating system was successfully deployed first, and only later understood as to why it

is effective. The contribution of this paper lies not in "inventing" information isomorphism, but in explicitly naming and systematically demonstrating this law for the first time, thereby completing the epistemological closure of the operating system.

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Keywords: Information Isomorphism Law; ontological law; epistemological meta-law; regime switching; statistical structure; Scientific Discovery Operating System

1. Introduction

1.1 An Unasked Epistemological Question

Tang (2026g, 2026r) established the theoretical framework of the Scientific Discovery Operating System. This operating system consists of three core components: the Factor Hierarchy Law (ontology) reveals how the world is organized—the hierarchical structure of regime factors and execution factors; the Testability Principle (epistemological standard) stipulates what counts as reliable—any claim must pass regime-dependence testing; the Testability Norms (methodology) prescribe how to operate—a standard framework of six functional dimensions (Search Completeness, Modulation Relationship, Structural Breakpoints, Causal Direction, Problem Completeness, Process Closure), within which specific implementation tools are replaceable. This operating system has been systematically verified across six completely independent disciplines: finance, astronomy, statistical physics, particle physics, chemistry, and nuclear physics (Tang, 2026a–2026ab).

However, a fundamental epistemological question has never been systematically asked: **Why is this operating system effective?** The Factor Hierarchy Law tells you "what to discover," the Testability Norms tell you "how to operate"—but neither answers "why operating in this way works."

This paper's answer is: because in systems that meet the relevant conditions, statistical structures carry the information of regime structures. This answer itself constitutes an independent law—the **Information Isomorphism Law**.

1.2 The Argumentative Path of This Paper

The argumentative path of this paper is not induction from specific discoveries to a universal principle, but a retracing from the operating system's success to its necessary premise. The six functional dimensions of the Testability Norms—Search Completeness, Modulation Relationship, Structural Breakpoints, Causal Direction, Problem Completeness, and Process Closure—together constitute a multi-dimensional probe of statistical structures. Search Completeness forces all candidate factors into the arena, Modulation Relationship tests interaction effects among factors, Structural Breakpoints detect systematic fractures in equation structures,

Causal Direction excludes endogeneity-based alternative explanations, Problem Completeness ensures comprehensive coverage of research questions, and Process Closure excludes methodological artifacts through negative controls. If statistical structures contained no information traces of regime switching, this methodology could not have succeeded repeatedly across six disciplines.

Therefore, the success of the operating system itself proves the establishment of the Information Isomorphism Law. The two precise dualities established on the 2D Ising exact solution provide precision verification for this law—they are not its inductive basis, but its precision test on a truth-known system. The inference from "operating system success" to "the Information Isomorphism Law is established" is not inductive generalization, but a necessary retracing of "result $\rightarrow$ premise"—if the premise did not hold, the result would be inexplicable.

1.3 Structure of This Paper

The structure of this paper revolves around the complete demonstration of the law: §2 provides the definition of the law and completes the core argument from the operating system's success to the law's necessity; §3 elaborates the three ontological levels of the law; §4 elaborates the complementary relationship between the law and the Factor Hierarchy Law; §5 discusses the self-reference of the law and its dual epistemological status; §6 elaborates the position of the law within the Scientific Discovery Operating System; §7 provides candid acknowledgments and testable predictions; §8 concludes.

2. Definition and Demonstration of the Information Isomorphism Law

2.1 Statement of the Law

The Information Isomorphism Law: In complex systems grounded in objective data and exhibiting regime switching, statistical structures carry the information of regime structures. This ontological fact is not an inductive generalization from specific discoveries to a universal principle, but a logical retracing from the success of the Scientific Discovery Operating System to its necessary premise—if statistical structures contained no information traces of regime switching, the six functional dimensions of the Testability Norms could not have succeeded repeatedly across six completely independent disciplines. On a truth-known system, the Chow F-order parameter duality and the intercept/slope ratio-critical distance duality provide precision verification for this law.

The Information Isomorphism Law is an **ontological law**—it asserts a fundamental fact about how the world is organized: "Statistical structures carry the information of regime structures." This is not a methodological prescription—"we should assume a correspondence between statistical structures and regime structures"—but an assertion about the world itself: "The world happens to be organized in this way." Precisely

because it describes the special fact of "information carrying," it is naturally and simultaneously an **epistemological meta-law**—it answers the fundamental question of "how we can know the world."

This definition requires layer-by-layer unfolding.

"Grounded in objective data": The observational data of the system are produced by automated equipment or natural forces, and their values do not depend on the subjective judgment of the observer. This is the sole applicability boundary of the operating system (Tang, 2026r, 2026z).

"Exhibiting regime switching": The system internally possesses at least one regime factor, and when the value of that factor crosses a critical threshold, the exposure coefficients of execution factors undergo systematic change. The existence of regime switching is described by Axiom 2 (Regime Switching) of the Factor Hierarchy Law.

The demonstrative logic of the law: The demonstrative logic of this paper is the necessary retracing of "result → premise." The six functional dimensions of the Testability Norms constitute a set of methodological tools. The core logic of this set of tools is: if statistical structures contained no information traces of regime switching, then exhaustive factor combinations would not automatically produce an optimal factor set (Search Completeness), the function of regime factors would not be concentrated in interaction effects rather than direct effects (Modulation Relationship), equation structures would not exhibit systematic fractures at specific positions (Structural Breakpoints), regime factors would not logically precede execution factors (Causal Direction), multiple negative controls could not simultaneously pass (Process Closure), and research questions could not be comprehensively covered across seven dimensions (Problem Completeness). These six dimensions are each independent—they capture information traces of regime switching in different dimensions of statistical structures. The systematic success of the operating system across six disciplines means that information traces on these six dimensions exist simultaneously across six disciplines—this is not a coincidence, but because in systems meeting the relevant conditions, regime switching necessarily leaves quantifiable traces in statistical structures.

The ontological content of the law: The Information Isomorphism Law describes the "information encoding mode" of the world—how regime switching is encoded into statistical structures. The first is **positional encoding**: the position of the regime boundary is encoded as the position of the statistical breakpoint—the Magic Number Shift Theorem (Tang, 2026v, Theorem 1) is the precise quantification of this encoding in nuclear physics. The second is **critical behavior encoding**: the continuous distance from the regime boundary is encoded as continuous variation of the statistical signature—the monotonic increase of the intercept/slope ratio with N in nuclear physics (Tang, 2026w, Theorem 9) and the systematic differentiation of switching types in particle physics (Tang, 2026o) are all convergent verifications of this encoding. Positional encoding and critical behavior encoding were not "induced" from the two dualities—they are encoding dimensions "extracted" from successful experience after the operating system was successfully deployed across

six disciplines. The two dualities provide precision verification for these two encodings on the 2D Ising exact solution, rather than serving as an inductive basis.

Precision verification: On the truth-known system (2D Ising exact solution), the Chow F-order parameter duality (Tang, 2026k) and the intercept/slope ratio-critical distance duality (Tang, 2026x, Theorem A) provide precision verification for the Information Isomorphism Law—they prove that on a truth-known system, information isomorphism holds precisely. It must be particularly noted that the functional positioning of the two precise dualities in the argument of this paper is not "inductive basis," but "precision verification." The establishment of the Information Isomorphism Law does not depend on the inductive generalization of these two dualities—it depends on the fundamental fact of the operating system's success across six disciplines. The value of the two dualities lies in proving that, on a truth-known system, this correspondence is not only existent but precise. It should be noted that what underwent rigorous calibration on the 2D Ising model are the two core tools of the Chow test and intercept/slope ratio analysis. Other structural breakpoint detection tools and activation operator stratifications within the operating system were primarily verified on the 3D Ising NP-complete system (Tang, 2026l), and have accumulated application verification through cross-disciplinary deployment across six disciplines.

The meaning of "Information Isomorphism": The term "isomorphism" in this paper does not refer to the strict mathematical sense of "a bijective mapping between two structures that preserves operations," but rather to "statistical quantities being precise information proxies for physical quantities"—statistical structures carry the systematic information of regime structures, making it possible to trace the attributes of regime switching through statistical tools.

2.2 The Intrinsic Connection Between the Law and the Success of the Operating System

The relationship between the Information Isomorphism Law and the success of the operating system is not "induction"—induction is the generalization from a limited number of successful cases to a universal law. The Information Isomorphism Law is a retracing from the fundamental fact of the operating system's success to its necessary premise. The logic of this retracing is: if the Information Isomorphism Law did not hold—that is, if statistical structures contained no information traces of regime switching—then the operating system could not have succeeded across six disciplines. Because every functional dimension of the operating system depends on the existence of information traces in statistical structures.

This is not circular reasoning. The success of the operating system is a fact independent of the Information Isomorphism Law—it has been repeatedly verified across six disciplines, and methodological artifacts have been excluded through multiple negative controls. The Information Isomorphism Law is the epistemological explanation of this fact. Facts precede explanations; explanations cannot substitute for facts, but explanations allow facts to be understood.

3. The Three Ontological Levels of the Law

The Information Isomorphism Law contains three progressive ontological levels. These three levels are not three independent facts discovered accidentally, but a logical unfolding from the single fundamental fact of "operating system success"—the success of the operating system across six disciplines requires these three levels to hold simultaneously, none being dispensable. The six functional dimensions of the Testability Norms depend on the Information Isomorphism Law at different levels.

The First Level: Existence (Information traces exist). Once regime switching occurs, it leaves quantifiable information traces in statistical structures. This is the most foundational level of the law—it answers the first fact of "how the world encodes information of regime switching": encoding necessarily occurs. The three core dimensions of Search Completeness, Modulation Relationship, and Structural Breakpoints in the Testability Norms directly depend on this level: if statistical structures contained no traces of regime switching whatsoever, exhaustive testing would not automatically produce an optimal factor set, interaction effects would not concentrate in regime factors rather than execution factors, and structural breakpoints would not systematically appear at specific positions. The success of these three core dimensions directly proves the establishment of the first level (existence).

The Second Level: Correspondence (Systematic correspondence exists between traces and regimes). The information traces left behind are not random, but exhibit quantifiable correspondence with the attributes of regime switching. This is the second ontological fact of the law: encoding is not random, but systematic. The Causal Direction dimension of the Testability Norms depends on this level: if information traces were merely random—that is, what regime switching left in statistical structures was merely irregular noise—then regime factors should not logically precede execution factors. The existence of precedence implies a certain systematic association between information traces and regime attributes—traces are not random, but are products of regime switching. Therefore, the success of Causal Direction verifies the "non-random association" between information traces and regime attributes—this is a necessary condition for "systematic correspondence." Positional encoding tells us "where the switching occurs" (the offset ΔN), and critical behavior encoding tells us "how far from the switching" (the magnitude of variation in the intercept/slope ratio r).

The Third Level: Precision (On a truth-known system, encoding is precise). On the 2D Ising exact solution, the Chow F-order parameter duality and the intercept/slope ratio-critical distance duality provide verification for the precision of this encoding. This is the third ontological fact of the law: encoding is not only existent and systematic, but also precise. The Process Closure dimension of the Testability Norms depends on this level: passing negative controls requires that information traces be not only existent but also statistically significant—if information traces were merely weak noise, multiple negative controls could not simultaneously pass. The success of Process Closure proves the establishment of the third level (precision).

The Intrinsic Unity of the Three Levels: Problem Completeness runs through all three levels—comprehensive coverage of seven-dimensional questioning requires that information traces hold simultaneously at all three levels of existence, correspondence, and precision. The three levels progress layer by layer, jointly supporting the success of the operating system across six disciplines. They are not "three levels discovered accidentally"—they are the three ontological conditions necessarily required by the single fact of the operating system's success.

4. The Complementary Relationship Between the Information Isomorphism Law and the Factor Hierarchy Law

The Information Isomorphism Law and the Factor Hierarchy Law are both meta-laws, describing two different aspects of the world, complementary rather than dependent.

The Factor Hierarchy Law (Tang, 2026r) describes the "organizational logic" of the world—complex systems possess a hierarchical structure of regime factors and execution factors; when a regime factor crosses a critical threshold, regime switching occurs, with two fundamental forms: rule-reset type and direction-reversal type. At the moment of its discovery, the Factor Hierarchy Law simultaneously revealed both an organizational principle of the world and the statistical visibility of that principle—it inherently carries an epistemological dimension. It is precisely for this reason that, after its discovery, it directly gave rise to the Testability Norms and the entire Scientific Discovery Operating System.

The Information Isomorphism Law describes the "information encoding mode" of the world—how regime switching leaves traces in statistical structures, and how statistical structures carry the information of regime structures. It is not an "epistemological supplement" to the Factor Hierarchy Law—the epistemological dimension inherent in the Factor Hierarchy Law was already sufficient to support the initial deployment of the operating system. The unique value of the Information Isomorphism Law lies in this: it separates "visibility" from the specific content of the Factor Hierarchy Law and establishes it as a universal law spanning all disciplines—"statistical structures carry the information of regime structures" is not merely an attribute of regime factors, but a universal attribute of all regime switching across all statistical dimensions. The Factor Hierarchy Law tells humanity that "the statistical signature of a regime factor is the interaction effect"—this is a specific visibility existing in a particular statistical dimension (Modulation Relationship). The Information Isomorphism Law tells humanity that "regime switching leaves information traces across all dimensions"—this is a universal visibility spanning all six functional dimensions.

The relationship between the two is complementary rather than dependent. The Information Isomorphism Law is not the "logical premise" of the Factor Hierarchy Law—the Factor Hierarchy Law was not deduced from the Information Isomorphism Law. Historically, the operating system was successfully deployed first (Tang,

2026a–2026f), and only later was it understood why it is effective (the Information Isomorphism Law being explicitly formulated in this paper). This is analogous to humanity using fire for hundreds of thousands of years before Lavoisier discovered the chemical principle of combustion. Fire kept burning not because humans understood oxidation reactions; similarly, the Factor Hierarchy Law kept functioning effectively not because anyone had theoretically demonstrated the Information Isomorphism Law. Yet the establishment of the Information Isomorphism Law is not dispensable—it upgrades the effectiveness of the operating system from "empirically repeatedly verified but not understood why" to "empirically repeatedly verified and understood why."

5. Self-Reference and Dual Epistemological Status

5.1 Self-Reference Analysis

The Factor Hierarchy Law describes "the organizational laws of complex systems"—this is an ontological proposition, which can be tested by itself. The Information Isomorphism Law asserts that "statistical structures carry the information of regime structures"—this is likewise an ontological proposition, and can likewise be tested by itself. The assertion of the Information Isomorphism Law is itself a multi-factor system—the operating system's own data (verification records across six disciplines) are produced by automated procedures, satisfying the condition of objective data; the regime factors within the operating system ("whether the data are objective") and execution factors (the testing procedures themselves) constitute a hierarchical structure. Therefore, the Information Isomorphism Law can be tested by itself—it satisfies its own conditions of applicability.

5.2 Dual Epistemological Status

The Information Isomorphism Law is a dual law spanning ontology and epistemology.

As an **ontological law**, it asserts a fundamental fact about how the world is organized: "Statistical structures carry the information of regime structures." This is not a methodological prescription, but an assertion about the world itself. The Factor Hierarchy Law describes regime switching itself; the Information Isomorphism Law describes the information encoding mode of regime switching. Both are ontological laws, describing two different aspects of the world.

As an **epistemological meta-law**, it answers the fundamental question of "how we can know the world." Precisely because it describes the special fact of "information carrying," it naturally possesses epistemological significance—it tells us why the attributes of regime switching can be traced through statistical tools, and why regime factors and execution factors can be identified through the probing of six functional dimensions.

6. The Position of the Information Isomorphism Law Within the Scientific Discovery Operating System

Tang (2026r) established the three-component structure of the operating system: the Factor Hierarchy Law (ontology), the Testability Principle (epistemological standard), and the Testability Norms (methodology). The Information Isomorphism Law supplements this structure with the missing foundation, completing a comprehensive four-layer structure for the operating system:

First Layer: The Information Isomorphism Law (Epistemological Foundation)—answers "why this methodology is effective." It establishes the ontological fact that statistical structures carry the information of regime structures, providing epistemological legitimacy for the operating system.

Second Layer: The Factor Hierarchy Law (Ontology)—answers "how the world is organized." It reveals the hierarchical structure of regime factors and execution factors, and the fundamental laws of regime switching.

Third Layer: The Testability Principle (Epistemological Standard)—answers "what counts as reliable." It stipulates that any claim must pass complete regime-dependence testing.

Fourth Layer: The Testability Norms (Methodology)—answers "how to operate." It provides the standard framework of six functional dimensions (with replaceable implementation tools).

The four layers progress hierarchically: epistemological foundation → ontology → epistemological standard → methodology.

7. Candid Acknowledgments and Testable Predictions

7.1 Candid Acknowledgments

Positioning of the Law: The Information Isomorphism Law is an ontological law, simultaneously possessing the status of an epistemological meta-law. Together with the Factor Hierarchy Law, they are both meta-laws, describing two different aspects of the world—the former describes the "information encoding mode," the latter describes the "organizational logic." The Information Isomorphism Law is not the "logical premise" of the Factor Hierarchy Law—the Factor Hierarchy Law had already been successfully deployed and verified before the Information Isomorphism Law was proposed.

Argumentative Path: The argumentative path of this paper is a retracing from the operating system's success to its necessary premise, not induction from limited instances to a universal principle. The Information Isomorphism Law is not a hypothesis "induced" from two dualities, but the necessary premise required by the operating system's success—if statistical structures contained no information traces of regime switching, the operating system could not have succeeded across six disciplines.

Delimitation of the Meaning of "Isomorphism": The term "isomorphism" in this paper does not refer to the strict mathematical sense of "a bijective mapping between two structures that preserves operations," but rather to "statistical structures being information proxies for regime structures"—statistical structures carry the systematic information of regime structures.

Testability: The law is testable. Its falsification conditions have been clearly stated in §2.4.

Calibration Scope: What underwent rigorous calibration on the 2D Ising exact solution are the two core tools of the Chow test and intercept/slope ratio analysis. Other structural breakpoint detection tools and activation operator stratifications within the operating system were primarily verified on the 3D Ising NP-complete system, and have accumulated application verification through cross-disciplinary deployment across six disciplines. The two precise dualities provide precision verification for the Information Isomorphism Law, not that "all tools have passed calibration on a truth-known system."

Scope of Precision: The fact that the Chow test has "strictly zero deviation" on the 2D Ising exact solution is a specific property of that tool on that system. Different statistical tools perform differently on the 2D Ising model according to their distinct mathematical principles.

Empirical Basis: All currently known verification instances come from a series of preprints by the same author and have not yet undergone independent third-party replication. Cross-disciplinary deployment across six disciplines constitutes convergent verification, but the universality of the law requires verification by more independent researchers.

Historical Order and Logical Order: The Information Isomorphism Law was explicitly proposed only after the completion of the preceding 28 papers. The contribution of this paper lies not in "inventing" information isomorphism, but in explicitly naming and systematically demonstrating this law for the first time, thereby completing the epistemological closure of the operating system. Historically, the operating system was successfully deployed first, and only later understood as to why it is effective.

Known Boundary Conditions: When data noise levels exceed the sensitivity of current tools, information traces, though existent, are undetectable. If regime switching is entirely continuous (without a critical threshold), information traces may manifest as trend drift rather than structural fracture. On truth-unknown systems, the initial judgment of "whether regime switching exists" may itself depend on statistical tools—this introduces a cycle that must be candidly acknowledged: we use statistical tools to detect regime switching, and the Information Isomorphism Law claims that statistical tools can detect regime switching. The method for breaking this cycle is to calibrate tools on a truth-known system, ensuring that the reliability of the tools does not depend on "our already knowing that regime switching is there."

7.2 Testable Predictions

Prediction One (Cross-Disciplinary Verification of the Operating System): In a seventh discipline not yet deployed, deploy the six functional dimensions of the Testability Norms. If core statistical tools detect significant structural anomalies at known or independently confirmed regime switching locations, this constitutes further convergent evidence for the Information Isomorphism Law. Falsification condition: if the six functional dimensions of the Testability Norms completely fail to detect any significant signal of known regime switching in the seventh discipline, the universality of the operating system is restricted to the six disciplines already deployed.

Prediction Two (Tool Independence): If a structural breakpoint detection tool based on entirely new mathematical principles is invented in the future, that tool should detect significant structural anomalies near T_c on the 2D Ising exact solution. Different tools will perform differently according to their distinct mathematical principles, but their convergent indication constitutes cross-tool verification of information isomorphism. Falsification condition: if the new tool detects no significant anomaly whatsoever near T_c .

Prediction Three (Cross-Disciplinary Validity of Information Dimensions): The currently known positional encoding and critical behavior encoding should have corresponding statistical quantities in the seventh discipline. Falsification condition: if, in the seventh discipline, positional encoding or critical behavior encoding is entirely absent.

8. Conclusion

This paper has proposed the Information Isomorphism Law: in complex systems grounded in objective data and exhibiting regime switching, statistical structures carry the information of regime structures. This ontological fact is not an inductive generalization from specific discoveries to a universal principle, but a logical retracing from the success of the Scientific Discovery Operating System to its necessary premise—if statistical structures contained no information traces of regime switching, the six functional dimensions of the Testability Norms could not have succeeded repeatedly across six completely independent disciplines. The Chow F-order parameter duality and the intercept/slope ratio-critical distance duality provide precision verification for this law on a truth-known system; the 3D Ising double-peak fission discovery provides application verification on a truth-unknown system; and cross-disciplinary deployment across six disciplines provides convergent evidence. The three ontological levels of the law—existence, correspondence, and precision—are not three facts discovered accidentally, but the three conditions necessarily required by the single fact of the operating system's success.

The Information Isomorphism Law is a dual law spanning ontology and epistemology. As an ontological law, it describes the "information encoding mode" of the world—how regime switching is encoded into

statistical structures. As an epistemological meta-law, it answers the fundamental question of "how we can know the world." The Factor Hierarchy Law describes the "organizational logic" of the world; the Information Isomorphism Law describes the "information encoding mode" of the world. Both are meta-laws, complementary rather than dependent. Within the Scientific Discovery Operating System, the Information Isomorphism Law occupies the epistemological foundation layer, completing a comprehensive four-layer structure. This paper has completed the epistemological closure of the Scientific Discovery Operating System—it has answered the fundamental question implicitly present but never explicitly formulated in the preceding 28 papers: "Why is this methodology effective?"

Data Availability Statement

This paper is a theoretical outline and systematic integration paper. All empirical data come from the series of preprints Tang (2026a–2026ab). The data are completely public, and any researcher may independently reproduce all results.

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